In Java Domain , there 3 categories of subject to learn to become java Sub expert

(a) JAVA Language (Programming Language)
(b) JAVA Technologies (Software Technologies)
(c) JAVA Frameworks (Software Frameworks)

Programming Languages
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=> Programming languages are raw material of the Programming providing basic facilities that are required for App development
=>Programming languages define the syntaxes (rules) and semantics (structure)of the Programming by giving compilers and interpeters

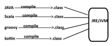
=>These are raw material of the Programming in application development (these are rice /wheat granuals)

=>These are base to create software technologies ,frameworks , operating systems, tools , DB s/ws and etc..

=>JAva language is BAse to create Andriod platorm (Mobile OS)
=> C language is base for creating unix ,Linux OS

eg:: c ,c++ , JAVA, vc++ , java script , python , C# , Ada , go, koltin , groovy ,scala and etc..

=>The languages whose compiler gives .class file after compilation and can be executed by using JVM is called JVM based languages



The .class files of JVM based languages can be executed by using the Common JRE/JVM

note:: There is no need of learning other JVM Based Languages as JAVA is also JVM based languages and it is continuously by getting new features of its own and inspired features other JVM based languages

Software Techologies
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=>These are software specifications providing set of rules and guidelines in the form of apis/libraries (packages having classes,interfaces and etc..) to create implementation softwares using one or another programming language

=> Software technologies are not installable , but the software technologies based implementation softwares are installable or arrangable. Working software technology implementation softwares is nothing but working with software technologies

eg:: JDBC , Servlet ,JSP , JNDI ,EJB ,JAAS ,JTA and etc..

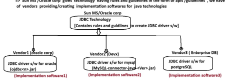
JSP :: Java SErver Pages
JNDI :: Java Naming Directory Interface
EJB :: Enterprise Java Beans
JTA :: Java Transaction API
JAAS :: Java Authentication And Authrorization Service

=> To work with JDBC technology we need to arrange JDBC driver s/w in the form of jar file
eg: JDBC driver s/w for oracle (ojdbc<n>.jar) , MySQL Connector/J Driver s/w (MySQL-connector-java-<ver>.jar)

=>To work with SErvlet Technology , we need to arrange SErvletContainer s/w (part of Web SErver s/w)
=>To work with JSP Technology , we need to arrange JspContainer s/w (part of Web SErver s/w)
=>To work with EJB Technology , we need to arrange EJB container s/w (part of Application SErver s/w)

Application server = web server++
⇩ weblogic , wildfly, etc.. ⇩ tomcat , jetty , undertow and etc..

=> Sun MS /Oracle corp gives technology having rules and guidelines in the form of apis /guidelines , we have lot of vendors providing/creating implementation softwares for java technologies



=>The software company who create and release softwares to the market is called software Vendor company
eg:: Microsoft , IBM , Oracle Corp , apache , google, amazon and etc..

=> The software company who create projects for client organizations by using the software created by the Vendor companies is called Software Services company

eg:: CTS , Capgemini , TCS , wipro , Oracle India , Infosys , NTTData, Emphasis and etc...